

VOXELFLIGHT

**A drone video you can
explore in three dimensions.**

For missions where another flight
may not be possible.

**Single-Pass Drone Video to Accurate 3D
Model Generation System**



ACTUAL INPUT FRAME

SIH26158

National Technical Research Organisation

Team name: [To be added] Team ID: [To be added]

Single-pass scene reconstruction

Video and telemetry support reconstruction of visible terrain, structures, roads, vegetation and obstacles.

01 Read the flight



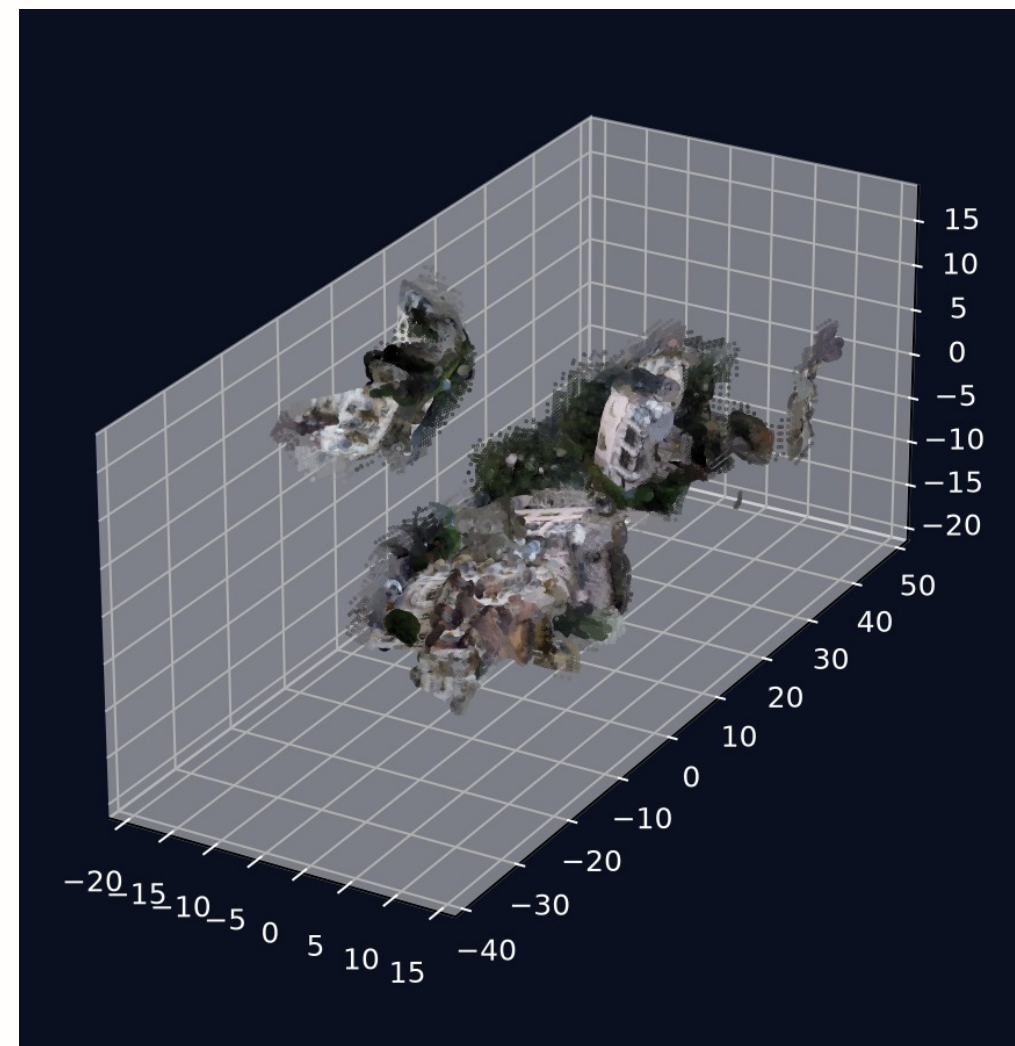
Video + GPS + flight metadata

Required video: 1080p or 4K

Recorded demo: 60 s, 180 keyframes

Optional: IMU, altitude, calibration,
RTK/PPK corrections.

02 Build geometry



Mesh / point cloud

Partial scene with fusion artifacts.

Spatial accuracy still needs validation.

03 Review appearance



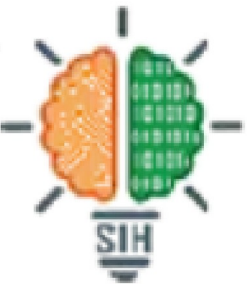
Optional Gaussian splatting

Fits the scene for realistic rendering.

Appearance cannot certify dimensions.

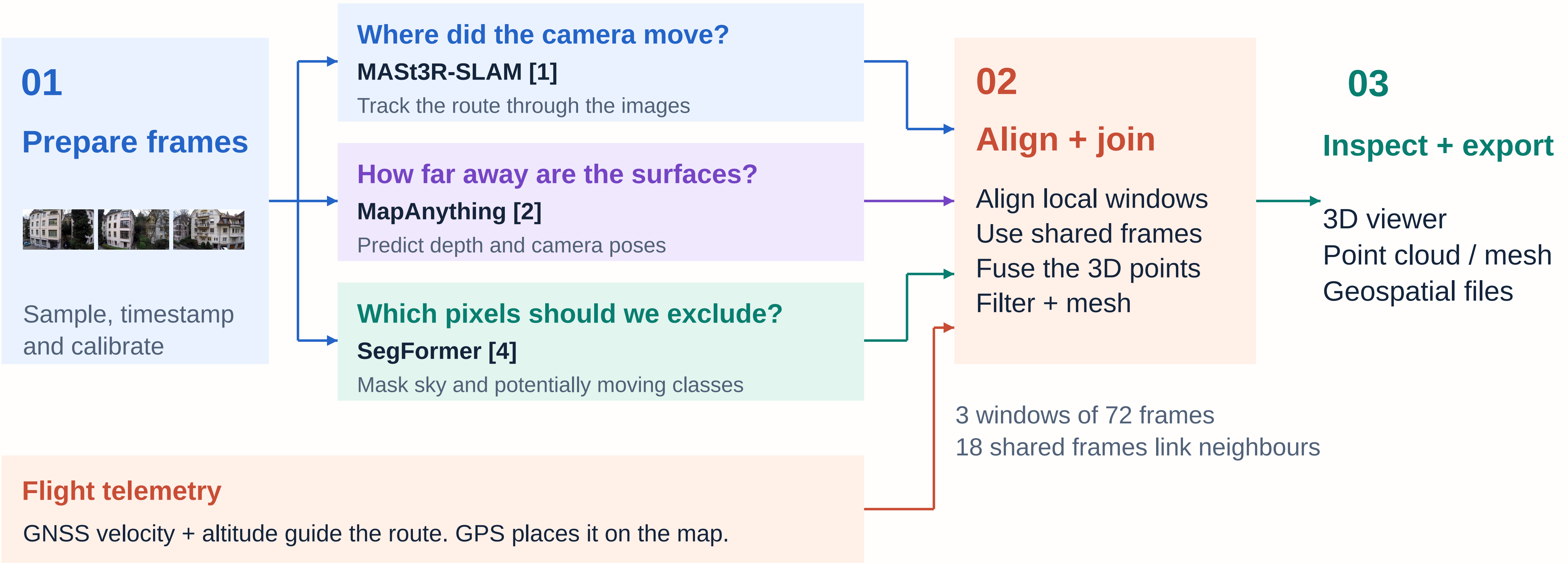
W000: PSNR 20.41 dB, SSIM 0.631

LPIPS 0.226

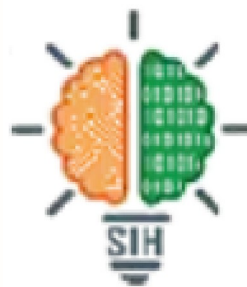


Multimodal reconstruction pipeline

Camera-motion estimates, depth predictions and flight telemetry combine in a shared scene coordinate system.



Cached weights. No ground-truth input to reconstruction. Optional 3DGS fits appearance separately.



Prototype performance and engineering strategy

Recorded reconstruction of a 60-second Zurich flight with 180 keyframes on an NVIDIA L40S GPU.

Measured property	Recorded result
Core reconstruction	3 min 29 s
With appearance refinement	5 min 34 s total
Clean point cloud	518,786 points
Exported mesh	450,000 triangles
Trajectory RMSE, Sim(3)-aligned	0.810 m
Trajectory RMSE, absolute	8.797 m

Geospatial output package

OBJ, PLY, LAS, GLB and GeoTIFF
Local ENU geometry with UTM-referenced exports

Limited views and scene motion

Overlapping windows retain shared observations.
Semantic masks exclude sky and potentially moving classes.

Metric scene placement

Visual anchors and GNSS/altitude constrain window placement. Reference poses enter evaluation only.

Execution and deployment

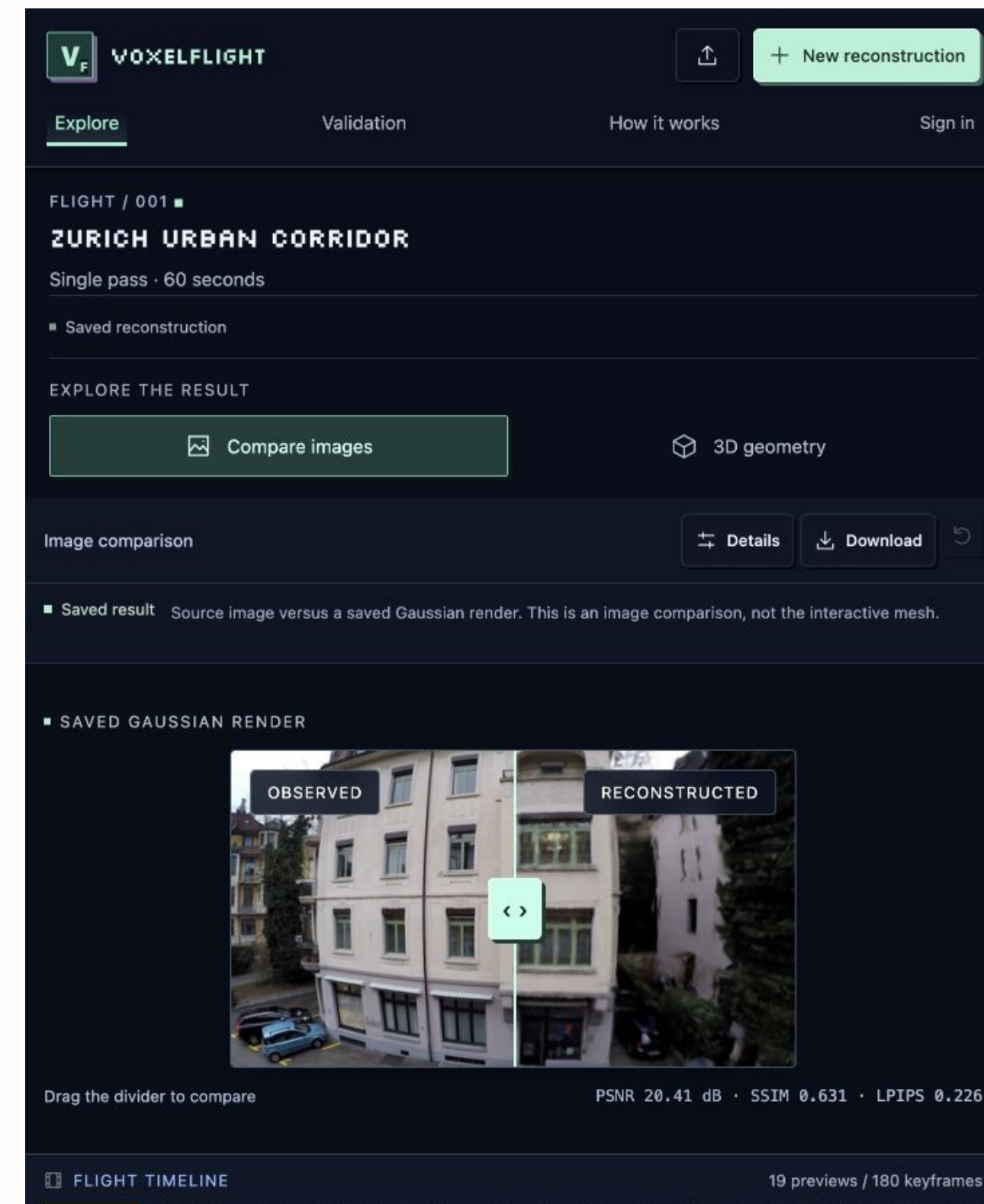
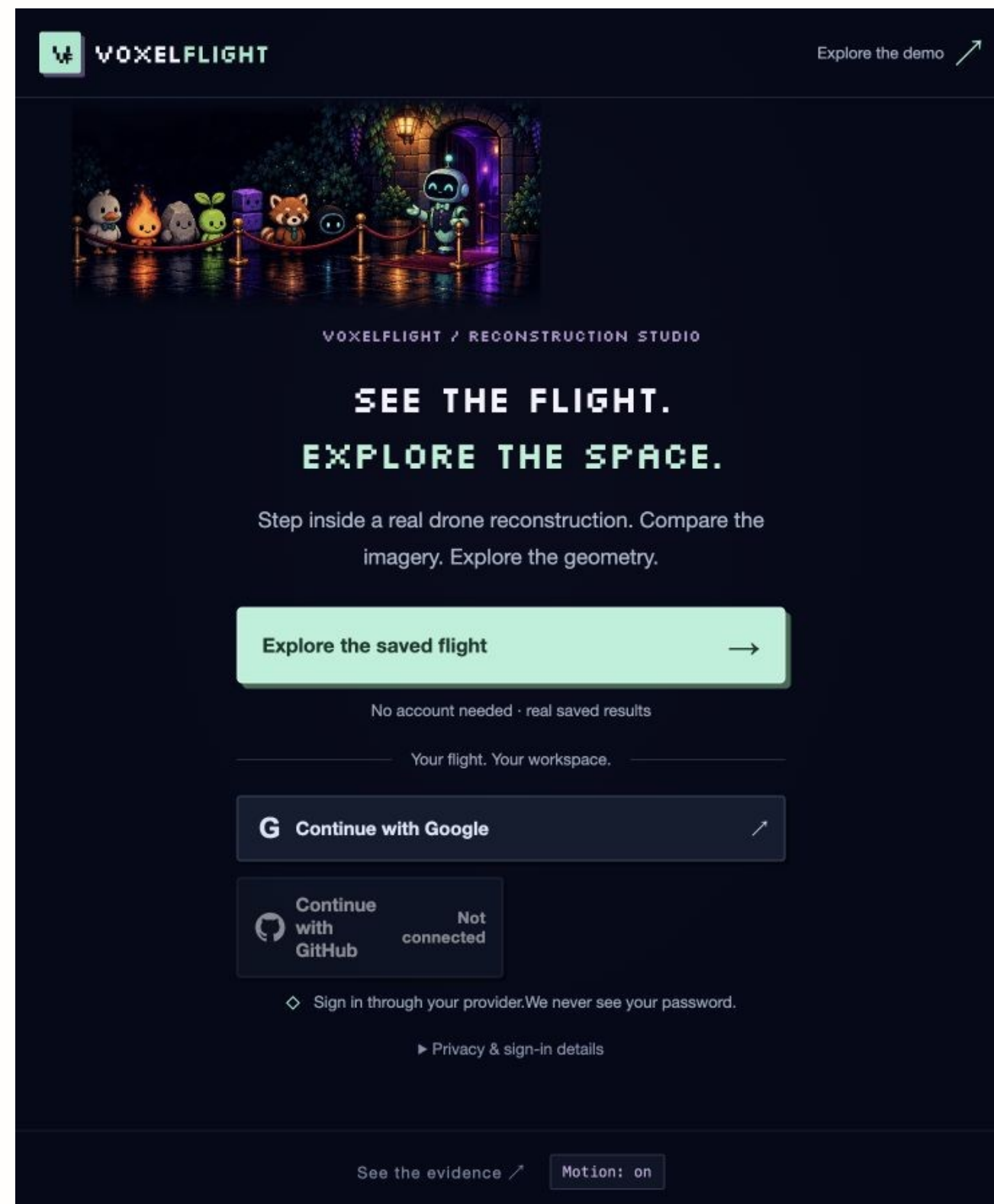
Cached model weights and bounded windows control the workload. A separate processing service supports the web interface.

Validation scope

Geometry is partial. Independent spatial accuracy and the 10-minute-video benchmark remain pending.
Trajectory RMSE does not measure surface accuracy.

Spatial context for inspection and planning

A browser workspace connects source imagery, reconstructed geometry and geospatial downloads.



Infrastructure inspection

Review façade and road context before a targeted follow-up survey.

Disaster assessment

Review visible damage remotely. Keep unseen regions explicit.

Planning and documentation

Reuse scene files with their coordinate reference information.

voxelflight-3d.web.app

Public prototype interface with saved reconstruction. Inference runs through a separate processing service.

Entrance + sign-in

Reconstruction workspace

Technical foundations and implementation contribution

Published reconstruction methods form the basis of the integrated VoxelFlight workflow.

[1] MAST3R-SLAM

CVPR 2025 / Visual camera tracking

[2] MapAnything

3DV 2026 / Feed-forward metric geometry

[3] 3D Gaussian Splatting

SIGGRAPH 2023 / Optional appearance refinement

[4] SegFormer

NeurIPS 2021 / Semantic segmentation and exclusion masks

[5] COLMAP

CVPR / ECCV 2016 / Classical reconstruction baseline

[6] Zurich Urban MAV

IJRR 2017 / Public imagery, telemetry and reference poses

Sensor-guided alignment

Combines visual motion, metric predictions and flight telemetry to place overlapping windows in a common coordinate system.

Geometry and appearance

Keeps mesh and point-cloud outputs separate from optional Gaussian rendering, with their respective validation results.

Connected geospatial delivery

Packages scene files with coordinate metadata and source imagery for browser inspection and downstream GIS workflows.

Technical focus: integrating complementary models with sensor-aware placement and inspectable scene outputs.